Maths Exploring the Terminal 11 Assessment

MCQ A · integral, area and medium

After S63; we always assume $a < b$.

For each question, only one proposal is accurate.
consult the correction.

1. If $F' = f$ on $[a;b]$, then $\int_a^b f(x)\,\mathrm{d}x$ is valid for:

A. $F(a) + F(b)$ B. $F(b) - F(a)$

C. $f(b) - f(a)$ D. 0

2. Integral of a negative function on $[a;b]$ is:

A. Positive B. None

C. Negative or nil D. Always a positive area

3. The average value of f on $[a;b]$ is:

A. $\frac{1}{b-a}\int_a^b f$ B. $\int_a^b f$

C. $f(a) + f(b)$ D. $b - a$

4. The geometric area between the curve and the axis of the abscisses requires:

A. to keep the signs B. cutting when the sign changes

C. a second derivative D. a random drawing

5. A antiderivative of e^x is:

A. xe^x B. e^x

C. $\ln x$ D. $1/e^x$

MCQ B · PPI and management

After S66.

For each question, only one proposal is accurate.
consult the correction.

6. For integration by parties, one must know:

A. Two grade functions $\mathcal{C}^1$ on $[a;b]$ B. a geometric sequence

C. a fixed point D. A binomial distribution

7. $\int_a^b u'v\,\mathrm{d}x$ is equal to:

A. $[uv]_a^b - \int_a^b uv'\,\mathrm{d}x$ B. $\int_a^b uv\,\mathrm{d}x$

C. $[u+v]_a^b$ D. 0

8. If f is increasing and positive on $[0;1]$, the rectangles on the left give :

A. a upper bound B. A minor

C. Always the exact integral D. a negative value

9. A complete sequence $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, f continued positive on $[0;1]$, may often be framed using:

A. $0 \leq x^n \leq 1$ B. $x^n = n$

C. $\ln x \geq 0$ D. the derivative of n

10. An integral flow unit (L/min) over a time in min is:

A. $\mathrm{L/min}^2$ B. L

C. min/L D. Ununited

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. An integral is always positive.

2. Two antiderivatives of the same function over an interval differ from one constant.

3. An approximation by rectangles does not control the direction of the frame.

Review review

I can interpret sign, area and medium; choose a antiderivative or an IPP and frame. **To be monitored:** confounding geometrical and integral algebraic area; forget the bounds in $[uv]_a^b$.